

No View from Nowhere

Read-out Closure and the Vacuum Curvature Test in Q-RAIF A

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Abstract

This paper states a predictive criterion in Q-RAIF A. Its central claim is that there is no complete observer-independent spacetime description: every spacetime description is conditional on an admissible read-out. This is not a claim about human subjectivity. It is a structural claim about closure: the read-out system cannot be placed completely inside the spacetime description that its own read-out makes available.

The cosmological constant problem is used as the sharpest test case. The question is not only how vacuum energy is prevented from gravitating, but why sector-internal zero-point bookkeeping was counted as a geometric source in the first place. The proposed read-out curvature criterion is that a vacuum contribution is eligible to source effective curvature only when it changes admissible read-out structure: causal accessibility, boundary or horizon conditions, modular/state-restriction data, stress response under admissible probes, or global read-out curvature conditions.

The framework therefore predicts that gravity couples to read-out-visible vacuum differences, not to absolute zero-point bookkeeping. Pure additive zero-point offsets inside a fixed admissible sector should not curve spacetime by their absolute value. Casimir energy, vacuum phase transitions, and global de Sitter horizon structure are different: they change boundary conditions, accessible modes, sector structure, or global read-out conditions. If a controlled vacuum offset that changes none of these structures nevertheless produces curvature proportional to its absolute zero-point value, the criterion fails.

Keywords: Q-RAIF A, read-out closure, cosmological constant problem, vacuum energy, emergent spacetime, zero-point energy, Casimir energy, observer independence

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1 The Blade

The sharp claim is this:

There is no complete observer-independent spacetime description. What physics calls spacetime is always conditional on an admissible read-out.

This is not idealism. It does not say that the human mind creates the world. It says that spacetime is not the primitive arena in which read-out occurs. In Q-RAIF A, effective Lorentzian geometry appears only when a finite accessible sector satisfies the admissibility constraints needed for Lorentzian read-out.

Equivalently, the claim is not observer-dependent spacetime in a psychological sense. It is read-out-conditioned spacetime in a structural sense. Spacetime is not produced by a mind; it becomes an admissible object only after the relevant read-out conditions close.

The usual picture places observers, measuring systems, records, detectors, horizons, and laboratories inside spacetime. That is correct after a read-out has been fixed. It is not correct as a fundamental starting point. Before the read-out is fixed, there is no complete spacetime container into which the read-out system can be inserted.

The structural error is closure failure:

A read-out system cannot be completely represented inside the read-out that it itself makes possible.

Many foundational puzzles become less mysterious when read this way. The measurement problem tries to place the measuring split inside a fully quantum description. The black-hole information problem asks for a single global account while using sector-dependent access boundaries. The cosmological measure problem asks for probabilities over global spacetime without a read-out-independent measure. The cosmological constant problem inserts sector-internal vacuum bookkeeping into geometry as if every internal term were already a spacetime source.

This paper does not attempt to solve all of these problems. It uses the cosmological constant as the most concrete test. If the read-out closure principle has physical content, it should say something falsifiable about vacuum energy. The other three problems are treated as stress tests in Section 6, not as solved problems.

2 Read-out Closure Criterion

Criterion 2.1 (Read-out closure). A physical description is complete only relative to a specified admissible read-out. A theory claiming a complete observer-independent spacetime must either:

1. introduce an explicit or hidden read-out structure;
2. fail to define records, probabilities, or classical sectors;
3. leave the emergence of spacetime underdetermined.

This is the general principle. The vacuum-energy version is sharper.

Criterion 2.2 (Read-out curvature). A vacuum contribution is eligible to source effective curvature only if it changes admissible read-out structure.

A vacuum contribution is read-out-visible if changing it changes at least one of the following:

1. causal accessibility;
2. boundary or horizon conditions;
3. modular or state-restriction data;
4. stress response under admissible probes;
5. global read-out curvature conditions.

These five entries are not an independent ad hoc list. They are the vacuum-energy specialization of the admissibility structure already used in Q-RAIF A:

Test item	Read-out meaning	Q-RAIF A source
A1. Causal accessibility	Which events, channels, or influences are distinguishable in the effective sector	Causal-channel distinguishability
A2. Boundary or horizon conditions	Which finite-access domain and boundary data define the sector	Finite-access domain / admissible boundary
A3. Modular or state-restriction data	How the state restricts to the accessible algebra and whether modular data change	State restriction / modular admissibility
A4. Stress response under admissible probes	Whether finite probes detect a changed stress or force response	Probe-readable observable response
A5. Global read-out curvature conditions	Whether the Lorentzian read-out remains globally consistent or changes horizon-scale structure	Global consistency of Lorentzian read-out

This gives a simple operational test.

Read-out Visibility Test. Fix an admissible sector $\mathcal{R}$ and its effective Lorentzian read-out. Change the proposed vacuum contribution while holding the sector fixed as far as the comparison permits. Check whether any of A1–A5 changes under admissible probes or boundary/global consistency conditions.

If at least one item changes, the contribution is read-out-visible and eligible to source curvature. If none changes, the contribution is sector-internal bookkeeping and is not eligible to source effective curvature by its absolute value.

Equivalently:

Gravity couples to read-out-visible vacuum differences, not to absolute zero-point bookkeeping.

The word “eligible” matters. The criterion does not compute the value of the observed cosmological constant. It says what kind of vacuum contribution is allowed to enter the geometry-source class in the first place.

3 The Vacuum Curvature Test

The standard cosmological constant problem begins with the semiclassical equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \langle T_{\mu\nu} \rangle.$$

For a Lorentz-invariant vacuum,

$$\langle T_{\mu\nu} \rangle_{\text{vac}} = -\rho_{\text{vac}} g_{\mu\nu},$$

so the vacuum energy looks like a contribution to the cosmological constant. A naive zero-point sum,

$$\rho_{\text{zp}} \sim \frac{1}{2} \sum_{\mathbf{k}} \hbar \omega_{\mathbf{k}},$$

then appears to predict an enormous curvature.

The read-out diagnosis cuts earlier. It asks whether this object is a geometric source at all.

A formal zero-point sum inside a fixed QFT sector may be meaningful as bookkeeping, regularization data, or renormalization structure. But if changing that offset does not alter causal access, boundary conditions, modular restriction, admissible stress response, or global read-out curvature, then it has not changed the effective spacetime read-out. It is not eligible to source curvature by its absolute value.

The issue is therefore not first how large the vacuum term is, but whether it has entered the class of geometric source terms at all. Eligibility precedes magnitude.

This is the prediction:

Pure sector-internal zero-point offsets do not gravitate naively.

This is not the weaker claim that a large value is cancelled. It is the stronger claim that the absolute bookkeeping term was never automatically in the geometric source class.

4 What Counts and What Does Not

The criterion gives a clean split.

Vacuum contribution	Read-out status	Curvature status
Constant shift $H \rightarrow H + C$ inside a fixed admissible sector	No change in read-out structure	Not eligible by absolute value
Formal bare zero-point sum	Regulator/bookkeeping object unless tied to read-out change	Not automatically eligible
Renormalization counterterm	Sector-internal unless tied to physical response	Not automatically eligible
Casimir energy	Changes boundary conditions, mode structure, stress response	Eligible
Vacuum phase transition	Changes spectra, condensates, effective masses, sector response	Eligible
Trace anomaly	Produces a physical renormalized stress response in curved spacetime	Eligible
de Sitter horizon scale	Changes global horizon/read-out conditions	Eligible

The Casimir case is the first sanity check. Casimir energy is not a pure absolute zero-point offset. It is a boundary-sensitive energy difference between configurations with different

allowed modes. It changes the accessible mode structure and produces stress on the boundary. Therefore it passes the read-out-visible test.

The same distinction applies to phase transitions. Electroweak symmetry breaking and QCD confinement are not merely new constants attached to the same sector. They alter excitation spectra, masses, condensates, and long-distance response. They can be read-out-visible.

The trace anomaly is another positive example. It is not a removable absolute zero-point offset; it is a physical renormalized stress response tied to curved-spacetime structure. In the present language it is read-out-visible through A4.

The observed cosmological constant, if positive, is also not naturally read here as a naive local sum of all zero-point modes. It is closer to a global read-out curvature condition, associated with horizon-scale structure.

The excluded class is narrow but decisive:

A vacuum term that changes no admissible read-out structure should not curve spacetime by its absolute value.

That is the knife edge.

This does not declare bookkeeping offsets unreal. It says only that a term may be real as sector bookkeeping while failing to change the read-out relations that define effective geometry.

The test can be applied explicitly.

Case	A1 causal	A2 boundary	A3 modular/state	A4 stress	Class
$H \rightarrow H + C$ in a fixed sector	No	No	Phase only	No	Offset
Bare zero-point sum with fixed sector	No	No	No read-out change	No	Not eligible
Casimir plate separation change	Cone unchanged; modes change	Yes	Restricted modes	Yes	Visible
Vacuum phase transition	Spectrum / channels	Possible	Yes	Yes	Visible
Trace anomaly in curved spacetime	Context-dependent	No local boundary	Renorm. stress data	Yes	Visible
de Sitter horizon-scale condition	Yes	Yes	Horizon restriction	Yes	Visible

This table is not a dynamical calculation. It is a source-eligibility classification. The criterion decides whether a vacuum contribution may enter the geometry-source class, not how large the resulting curvature must be.

5 Relation to Existing Approaches

The proposal differs from several familiar strategies.

Unimodular gravity changes the role of the trace part of Einstein's equation and treats the cosmological constant as an integration constant.

Vacuum energy sequestering introduces global constraints so that vacuum energy does not gravitate in the usual way.

Degravitation modifies the gravitational response to long-wavelength sources.

Induced gravity and related vacuum-response programmes try to obtain gravitational dynamics from quantum vacuum or matter-sector response.

Standard QFT in curved spacetime treats the renormalized stress-energy tensor carefully, including regularization dependence, trace anomaly, and state dependence. This paper does not replace that machinery. Its claim is prior: not every sector-internal vacuum bookkeeping term is automatically source-eligible for emergent curvature.

The physical distinction between measurable vacuum differences and absolute offsets is already implicit in axiomatic renormalization, especially in Wald's treatment of the renormalized stress-energy tensor and the cosmological-constant renormalization freedom. The present criterion's contribution is not to rediscover that bookkeeping point, but to ground it in a structural principle: read-out closure.

The read-out criterion asks a prior question:

Why was sector-internal vacuum bookkeeping counted as a geometric source in the first place?

It is not a cancellation mechanism. It is not an infrared filter. It is not a trace-rearrangement of the same source. It is a source-eligibility criterion arising from the claim that geometry is a read-out structure, not a container for every algebraic bookkeeping term.

Approach	Entry question	Difference from the read-out criterion
Unimodular / trace-free gravity	How should the trace or cosmological-constant part of the field equations be handled?	Assumes an effective geometric equation and changes its trace structure
Sequestering	How can vacuum energy be globally neutralized?	Treats vacuum energy as a source to be constrained or cancelled
Degravitation	How can gravity filter long-wavelength sources?	Modifies the gravitational response rather than the source-eligibility test
Induced gravity	Can geometry arise from quantum vacuum or matter response?	Uses vacuum response constructively; does not by itself separate bookkeeping offsets from read-out-visible differences
Read-out criterion	Why was sector-internal vacuum bookkeeping counted as a geometric source?	Tests source eligibility before inserting a term into emergent curvature

6 Three Wider Stress Tests

The same closure principle appears in three other problems, but here they are used only as stress tests.

Measurement. Many-worlds requires a branch structure. Copenhagen requires a classical apparatus. Decoherence requires a system/environment split. Wigner’s friend scenarios make the closure failure operational: the friend, the record, and the external observer cannot all be placed into a single completed read-out without specifying which read-out defines the record. None of these is a view from nowhere. Each uses a read-out structure to define records and outcomes.

This is related to, but distinct from, relational quantum mechanics. RQM relativizes facts to physical systems; the present claim is stronger in a different direction, because it says that the spacetime arena itself is conditional on admissible read-out. Frauchiger–Renner type no-go results sharpen the same pressure by showing that universal quantum theory, single outcomes, and cross-observer consistency cannot all be maintained without qualification.

Black-hole information. Modern Page-curve and island calculations are not read-out-free. They depend on how radiation, interior, exterior, island, and gravitational sectors are assigned. The question is not simply where information goes. It is which sector has access to which imprint.

Cosmological measure. In eternal inflation or multiverse settings, there is no obvious read-out-independent probability measure over all events. Measures introduce cutoffs, slicings, observer classes, or sampling rules. These are read-out choices in mathematical form.

AdS/CFT. AdS/CFT may appear to provide a complete observer-independent description through the boundary CFT. The present reading treats it differently: the boundary state, operator content, and boundary conditions specify the data from which bulk geometry is reconstructed. They therefore function as read-out data rather than as a view from nowhere.

These examples are not claimed as solved. They show the same pattern:

A theory that claims total observer independence must still smuggle in a structure that selects records, sectors, probabilities, or access.

7 Predictions and Falsifiers

The paper’s predictions are exclusionary but sharp.

P1. Absolute zero-point non-gravitation. Pure additive zero-point offsets inside a fixed admissible sector should not produce curvature proportional to their absolute value.

P2. Boundary-sensitive vacuum gravitation. Vacuum differences that change boundary conditions, accessible modes, or stress response, such as Casimir-type configurations, remain eligible to gravitate.

P3. Phase-transition eligibility. Vacuum phase changes can enter geometry because they change the read-out sector, not because every bare vacuum number is automatically a source.

P4. No complete view from nowhere. As a structural expectation, any complete quantum-gravity account of records, probabilities, horizons, and cosmological measures should contain an explicit or hidden read-out/sector-selection structure.

The clean falsifier is:

If a controlled vacuum-energy shift that changes no admissible read-out structure nevertheless produces curvature proportional to its absolute zero-point value, the read-out curvature criterion fails.

The broader falsifier is:

If a complete observer-independent quantum-gravity theory defines spacetime, records, Born probabilities, black-hole information accounting, and cosmological measure without any primitive or emergent read-out/sector-selection structure, the read-out closure criterion fails.

A nearer-term challenge would be a specific non-perturbative quantum-gravity model, such as a complete string vacuum or comparable construction, that defines bulk spacetime, Born probabilities, and entropy accounting without boundary, state, sector, cutoff, branch, or record-selection data.

For this to count as a failure, the absence of read-out structure must be independently identifiable, not achieved by retroactively declaring every needed sector split, branch rule, cutoff, or record map to be “not a read-out.” The criterion is falsified by a genuinely closed construction, not by a change of vocabulary.

These are demanding tests, but they are not vague. The framework predicts that the “view from nowhere” will not close.

8 Limits

This paper does not compute the observed cosmological constant. It does not replace semiclassical gravity. It does not deny that Casimir energy, phase transitions, or global vacuum structure can have gravitational significance.

The criterion removes the bookkeeping class from the source. It does not explain why the remaining read-out-visible contributions, such as Higgs vacuum structure, QCD condensates, or phase-transition vacuum shifts, do not produce curvature at their naive scale. That is a separate problem, possibly requiring dynamical cancellation, sequestering of physical contributions, or a deeper read-out mechanism not supplied in this note.

It also does not identify read-out with human consciousness. The read-out is a structural condition for an admissible physical description, not a psychological act.

In this sense, the slogan “no view from nowhere” should not be read as a subjectivist claim. It means that there is no completed spacetime arena prior to the conditions that make spacetime readable.

The minimal claim is:

1. Spacetime descriptions are conditional on admissible read-out.

2. A read-out system cannot be fully placed inside its own read-out without hidden structure.
3. Vacuum energy sources curvature only when it changes read-out-visible structure.
4. Pure absolute zero-point bookkeeping is not automatically a source of geometry.

That is the intended sharp point:

The universe has no complete spacetime view from nowhere; geometry begins only where read-out closes.

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